

ANURAG KADKOL

EDUCATION

Northeastern University
Khoury College of Computer Sciences
MS Robotics (CS)

09, 2024 - Present
Boston, MA

BITS Pilani
Dual Degree Program

08, 2018 – 06, 2023
Goa, India

MSc Biological Sciences
BE Electronics & Communication Engineering

PUBLICATIONS

1. **Kadkol A.**, Bhati R., Baisero A., Amato C., "Decentralized Asymmetric DQN". *RLC 2026, in preparation.*

EXPERIENCE

Northeastern University, USA
Graduate Teaching Assistant

09, 2025 – Present

Course: Reinforcement Learning & Sequential Decision Making

- Conduct weekly office hours to support student understanding of RL concepts and assignments.
- Grade assignments and exams for a class of over 150 students.
- Collaborate with the instructor to design exams and assessment materials.

Northeastern University, USA
Graduate Student Researcher

09, 2025 – Present

Project: LLM Assisted Reinforcement Learning

Advisor: Dr. Lawson Wong

- Developing Advice Guided RL framework that uses LLMs to generate exploration policies in RLang (a MDP specification DSL), with current results demonstrating faster convergence and higher performance than PPO baselines in Minigrid and OCArari environments.
- Designing adaptive mixing schemes for optimal integration of advice into RL training pipelines.
- Leveraging LLM generated RLang advice policies to generate synthetic datasets for behavior cloning, enabling PPO initialization with optimistic priors to accelerate learning.

Northeastern University, USA
Graduate Student Researcher

03, 2025 – Present

Project: Decentralized Asymmetric Deep Q Networks

Advisor: Dr. Christopher Amato

- Extended an asymmetric reinforcement learning algorithm (ADQN) to multi-agent settings, developing a novel MARL framework that achieves decentralized execution without value factorization, resulting in richer cooperative behavior.
- Implemented and benchmarked MARL baselines (IQL, VDN, QMIX, QPLEX) for comprehensive evaluation, with the proposed method outperforming all baselines on SMAX and Overcooked environments.
- Currently extending framework to continuous control domains using Cross Entropy Method and developing theoretical guarantees to explain empirical success.

CWI, Netherlands**08, 2023 – 12, 2023***Research Intern***Project:** Programmatic Reinforcement Learning using LLMs**Advisor:** Dr. Hendrik Baier

- Explored programmatic reinforcement learning in the MicroRTS environment, focusing on policy generation using Large Language Models in the Microlanguage DSL.
- Worked on using open-source LLMs(Llama2) to generate strong initial candidate policies for local search algorithms.
- Enabled model usage on consumer-grade GPUs through 4-bit quantization techniques, eliminating dependency on expensive cloud infrastructure for programmatic policy generation.

Uchida Lab, Harvard University, USA**01, 2023 – 08, 2024***Research Intern***Project:** Distributional Reinforcement Learning in the Human Brain**Advisor:** Dr. Naoshige Uchida

- Compared representations in deep RL algorithms(DQN, C51, QRDQN) to neural representations in the human brain. Demonstrated that hidden layers of deep RL models predict neural responses throughout the Dorsal Visual stream and Posterior Parietal Cortex(PPC)
- Trained DQN and various distributional RL algorithms like Categorical DQN and Quantile Regression DQN.
- Validated alignment between the DQN policy and the human behavior policy by extracting action advantages for over 100,000 gameplay frames, comparing RL model values between chosen vs non-chosen actions, and showing higher values for participant-selected actions.
- Decoded human left/right actions from hidden layer activations using LASSO logistic regression on 400 PCA-reduced features validated through cross-validation across runs and block-permutation tests.
- Mapped model representations to distributed sensorimotor network using cross-validated ridge regression, showing significant prediction accuracy throughout the dorsal visual cortex and the PPC.

AIML Lab, TU Darmstadt, Germany**08, 2022 – 12,2022***Research Intern***Project:** Incorporating Attention Mechanism in Reinforcement Learning Algorithms**Advisor:** Dr. Kristian Kersting

- Designed novel Procgen environments and efficiently tuned their associated reward functions.
- Trained 'experts'/sub-policies that solved sub-tasks in the designed environments.
- Implemented an algorithm that divided a task into sub-tasks and selects 'experts' to solve the sub-tasks. The implemented algorithm achieved performance comparable to PPO while being explainable.

Nectorr Labs, Gurgaon, India**05, 2020 – 07,2020***Web Development Intern*

- Created a full-stack analytics web application that fetched trending keywords on various social media platforms using recurring calls to Twitter, YouTube and Google Trends APIs.
- Developed the backend and the database for the web application using Node.js and MongoDB.
- Secured the application with a login authentication system using Passport.js and Node.js.

PROJECTS

Solving Hex Using Self-Play PPO

- Developed a custom Hex game environment using C, Python, and Cython that runs at over 1.2 million steps per second.
- Engineered scripted opponents of increasing complexity, with the advanced bot utilizing virtual connections, bridge patterns, and ladder formations with look-ahead planning.
- Implemented Proximal Policy Optimization (PPO) to train agents against three difficulty-scaled opponents.
- Designed and evaluated four neural network architectures (FC, CNN, FC-LSTM, CNN-LSTM) for strategic gameplay, achieving 100% win rates against all opponent levels.

Synthetic Multimodal SLAM dataset generation

- Created diverse 3D environments using manual modeling (e.g., classrooms), satellite reconstruction with OpenStreetMap data, and procedural generation with Perlin noise and probabilistic sampling.
- Designed scalable indoor and outdoor scenes (forests, mountains, and cityscapes), to test SLAM robustness under varied layout and texture conditions.
- Generated trajectories using Bézier curves to simulate varied motion patterns for the virtual camera.
- Simulated RGB, depth (via Z-buffer), and LiDAR (via raycasting) sensors, varying camera parameters, and exporting ground-truth pose data for supervised learning.

Genomic Analysis of *Arabidopsis Thaliana*

- Reviewed the literature on stress regulating cis-regulatory elements in plants.
- Identified and studied biotic and abiotic stress regulatory motifs of *Arabidopsis thaliana*.
- Analysed distribution and spacer frequency of heat-shock elements across the entire genome.

Cross-Platform Game Porting and Localization

- Ported two of the four routes of a visual novel from PS Vita/PS4 to PC/Mac using renPy.
- Worked closely with developers and translators from across the world and translated the visual novel from Japanese to English.

COURSEWORK AND SKILLS

Courses: Robotics Science & System, Robotics Sensing & Navigation, Robot Mechanics and Control, Reinforcement Learning & Sequential Decision Making, Numerical Optimization Methods, Digital Image Processing, Digital Signal Processing, Control Systems, C Programming

Software and Programming: Python, Pytorch, JAX, Java, Unity, C#, C, C++, ROS, $\LaTeX$

LANGUAGES

English: Proficient (Professional working proficiency)

Japanese: N3 (Intermediate)

Marathi, Hindi, Kannada: Native